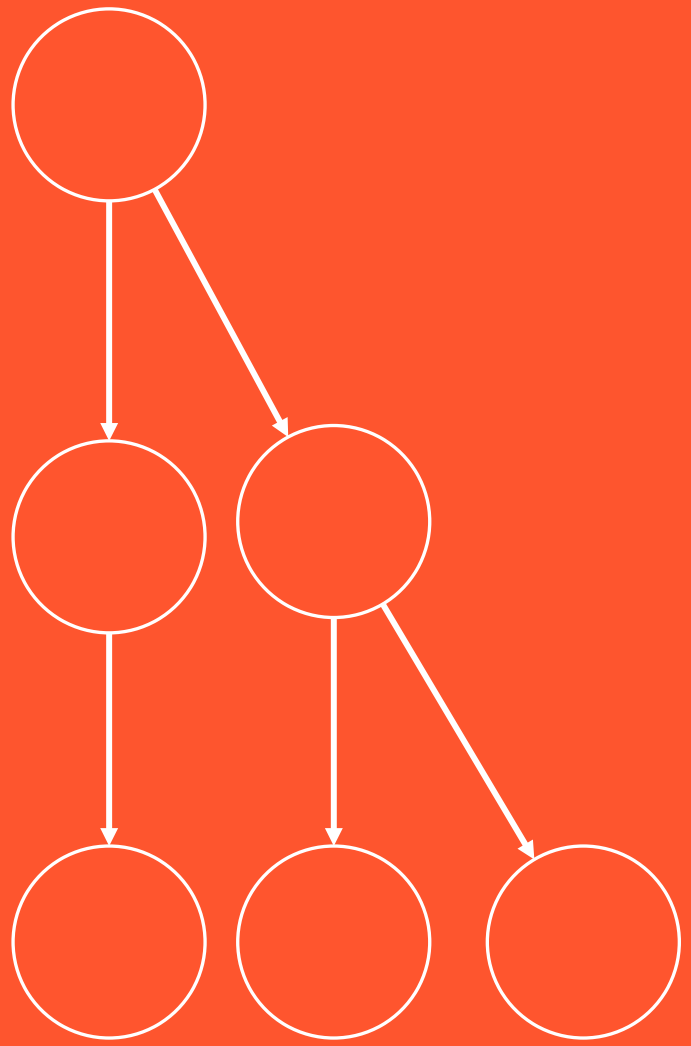


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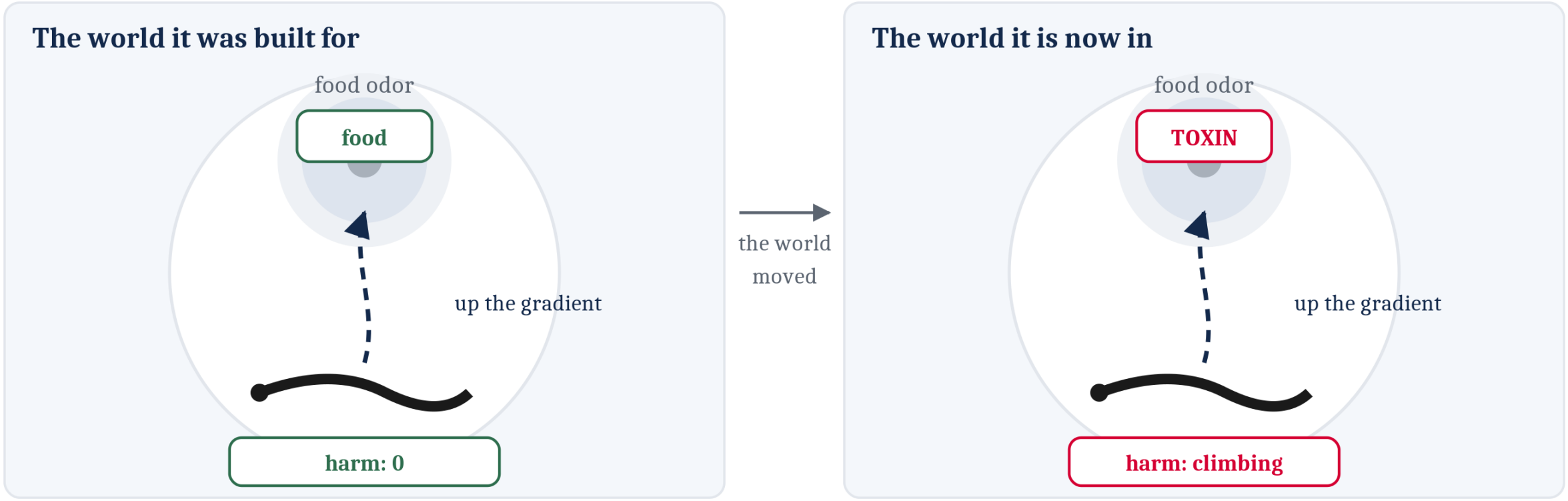
Learning and Plasticity

Lecture 5.1 — The Evolutionary Advantage of Learning

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Change the World, and a Fixed Animal Fails the Same Way Every Time

Move the food. Or let a smell that has always meant a meal come to mean poison. An animal whose **weights are fixed** goes on producing what it was built to produce, and fails identically every time.

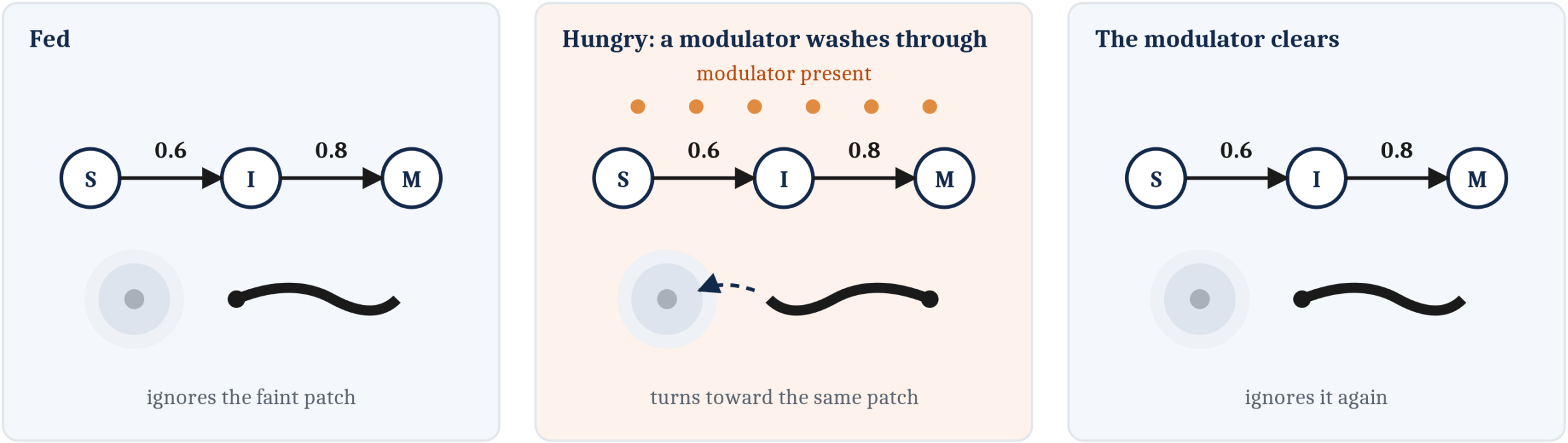


The animal did not. Its weights are the same numbers, so the behavior is the same behavior.

From outside, we can see which weight should change. Nothing in the animal can reach it.

Modulation Is Real Flexibility, and It Is Not Learning

A **weight** sets which input counts for how much, and with what sign. A neuromodulator changes what the same circuit does with the same input. When the chemistry clears, every weight is where it was.



Modulation changes what the wiring makes of an input. It keeps nothing.

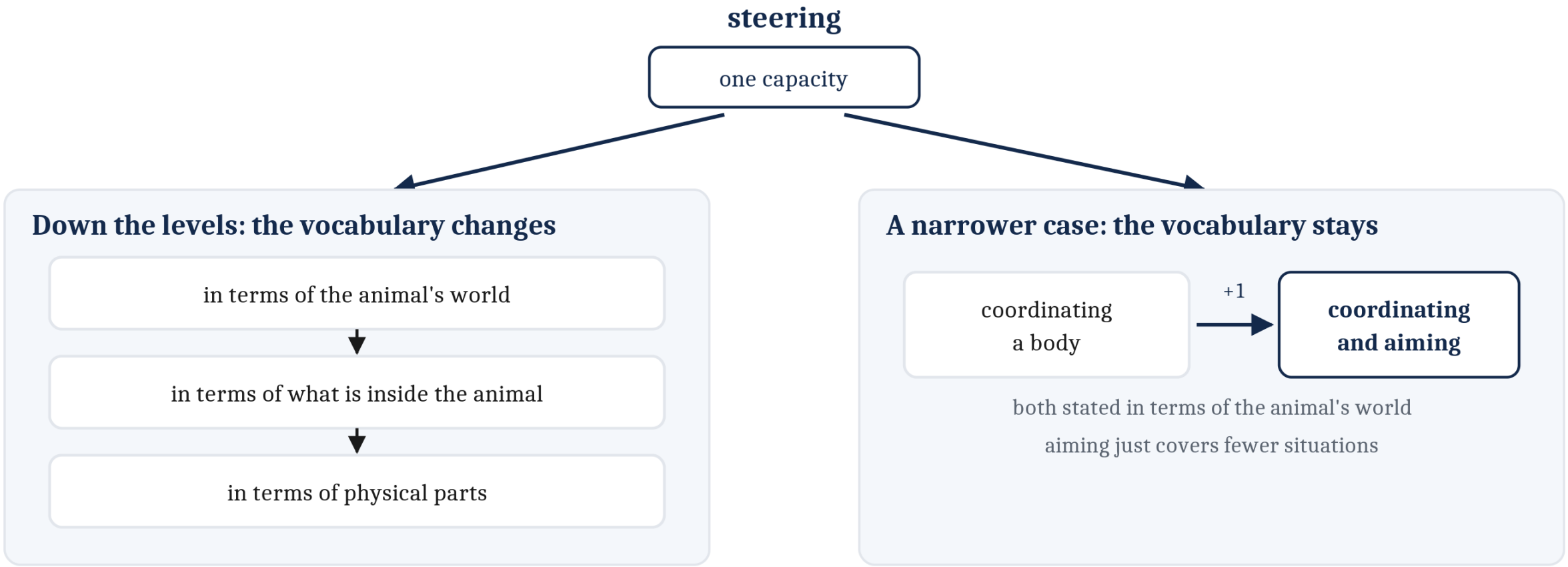
Marr's Wheel, a Third Time

Module 3 asked the computational question of a cell. Module 4 asked it of an arrangement of cells. This module asks it of a **change**.

Level	The question it asks	Discipline	This module's example: a weight that can change
Computational	What problem is being solved, and why does solving it matter?	Evolution and ecology	Detecting contingencies that selection could not have fixed a response to: ones that change within a lifetime, ones that differ from place to place, and ones that are new to the species
Algorithmic	What is represented, and by what procedure?	Cognitive psychology and AI	Rules that move a weight — on local activity alone, on a target supplied by the world or by a teacher, or on one broadcast number reporting how things went
Implementational	By what physical means, and at what cost?	Neuroscience	A synapse that detects coincidence, a signal from elsewhere that gates what the coincidence does, and a fading mark that decides which synapses are still candidates

Two Different Ways a Question Gets More Specific

Moving **down the levels** changes the kind of question, and the vocabulary with it. A **narrower case** keeps both and adds a constraint. It earns its own name when it changes what a solution must have.

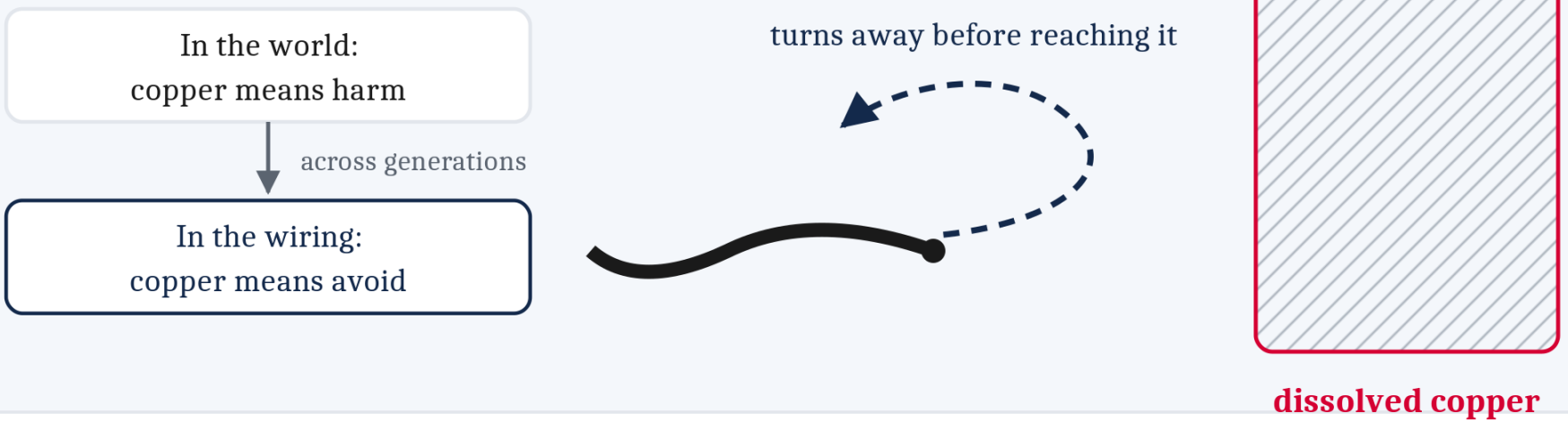


The test is which vocabulary the description is stated in.

A narrower case earns its own name when it changes what a solution must have.

Detecting a Contingency Is Not the New Thing

A **contingency** is a reliable relationship between one thing and another. The worm avoids copper, and nothing in its own life taught it to: that relationship was detected across generations.

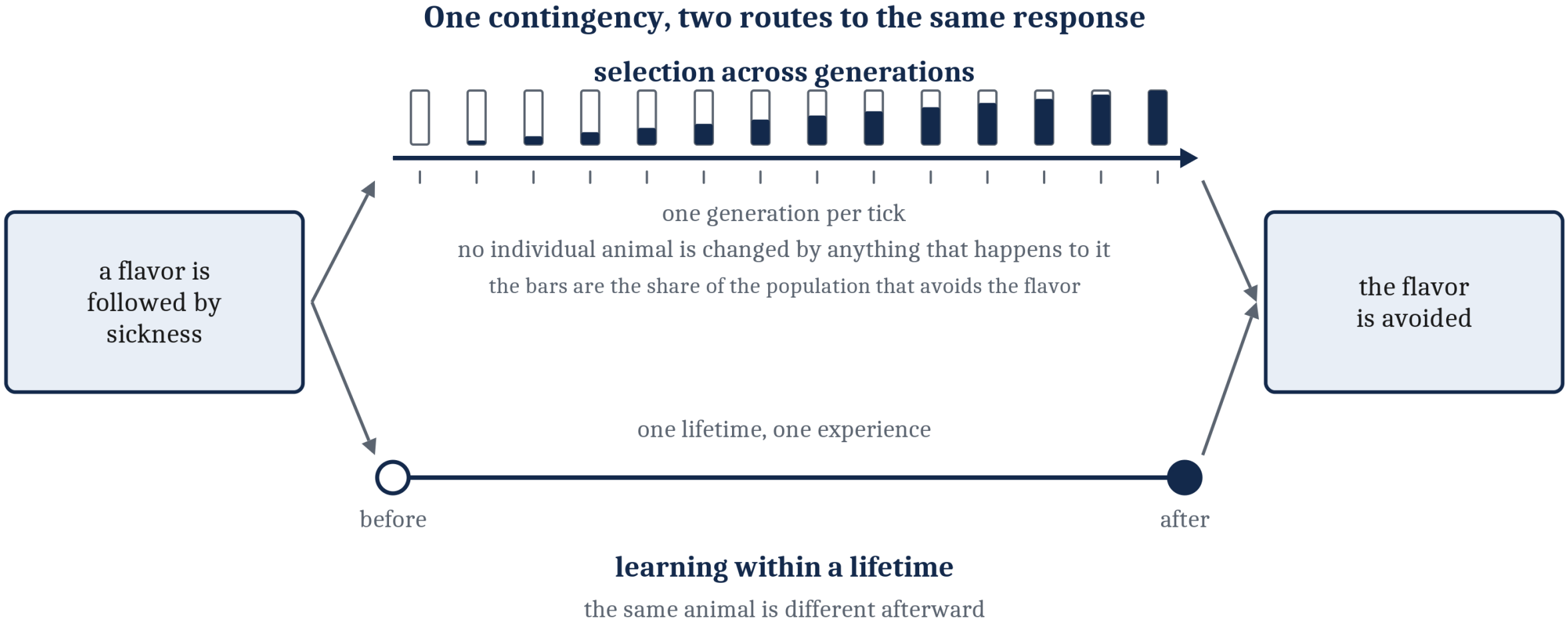


A worm reared without ever meeting copper avoids it on first contact.

Nothing was trying to do the fitting. Nothing had to know copper was dangerous.

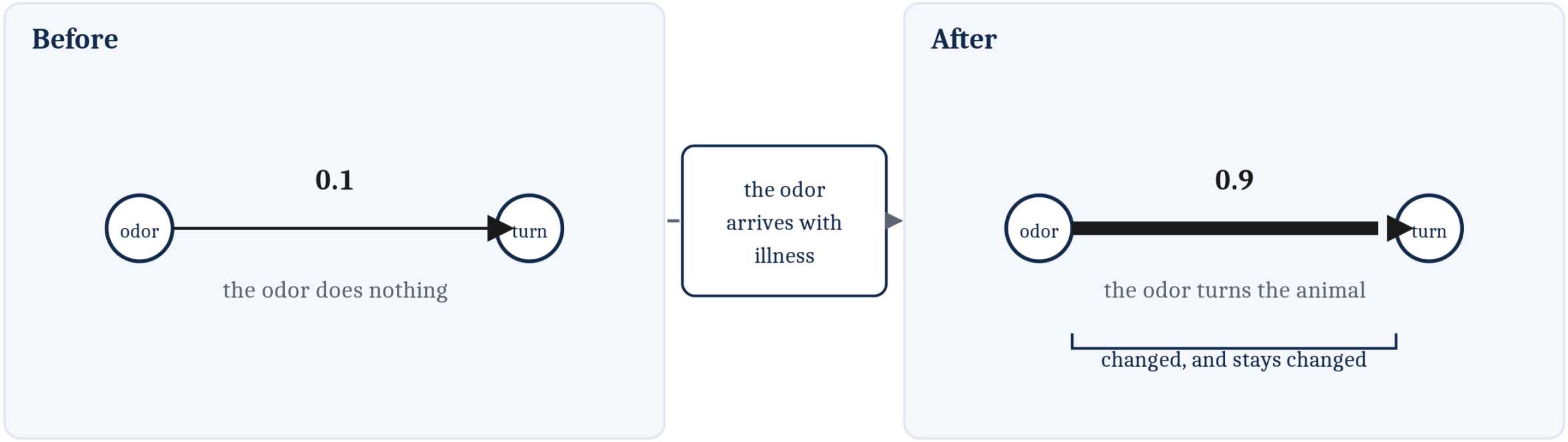
Selection Is Slow and Coarse, and Three Kinds of Relationship Slip Through

Contingencies that **reverse within a lifetime**, that **differ from place to place**, or that are **new to the species** cannot be met by a response fixed in advance. That is the module's problem.



What an Acquired Response Requires

Learning is a lasting change in what an animal does, produced by experience and resting on an internal state experience can modify. **Plasticity** is what makes such a state possible.



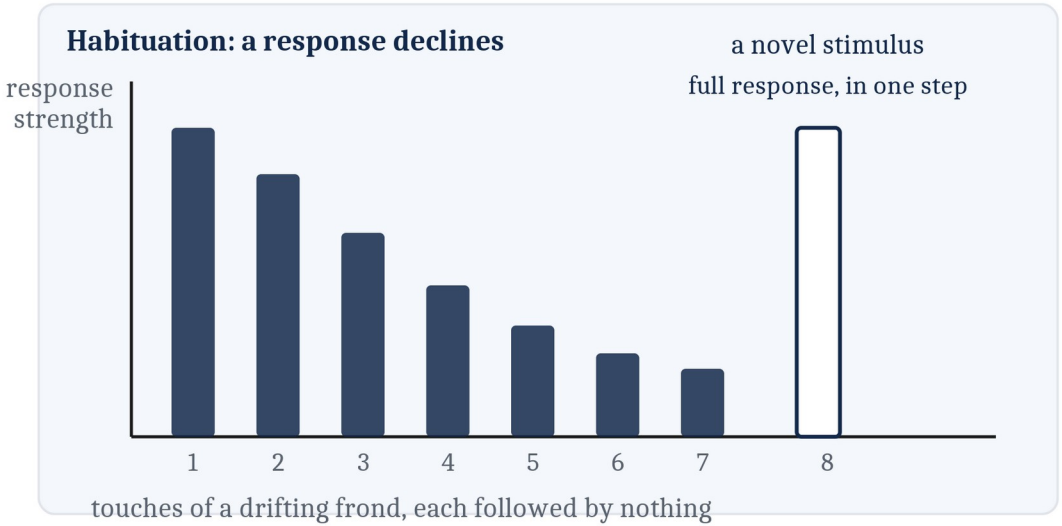
LEARNING: the lasting change in what the animal does

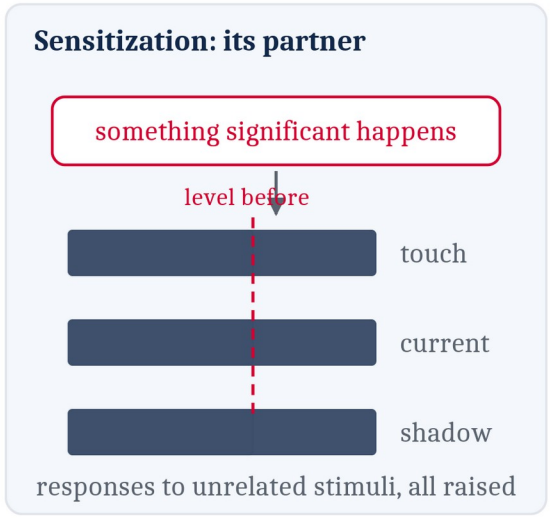
PLASTICITY: the tissue's capacity for such a change

Not a correction of something wrong. Nothing in the animal marked the old weight as a mistake.

Tier One: Changing the Response to Something That Keeps Happening

Habituation is a decline in a response to a stimulus repeated without consequence. **Sensitization** is its partner, a rise in responsiveness after something significant happens.







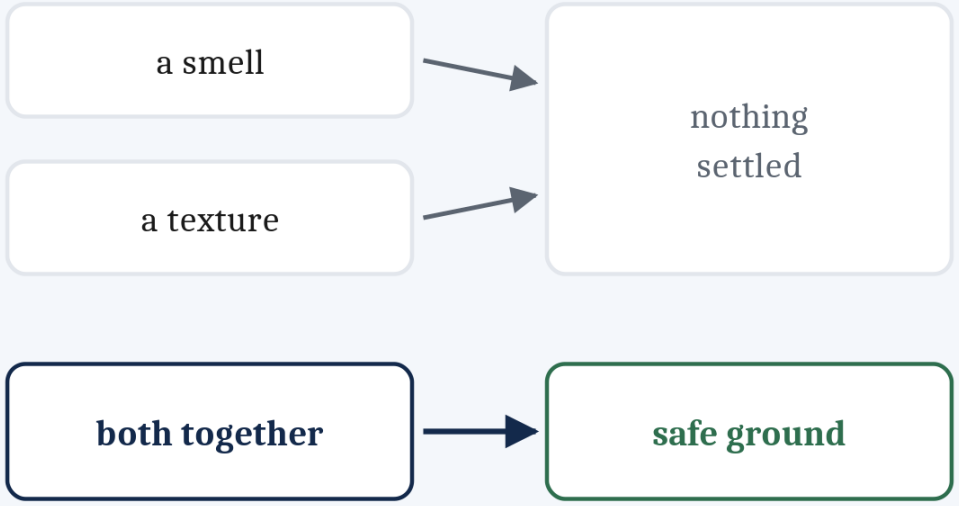
Nothing is used up, so this is not fatigue. Something has been adjusted.

Tier one needs no link between two events, and no outcome to compare against.

Tier Two: Linking Two Things That Occur Together

A link between two states buys two things: acting **before** an event, and treating a **novel combination** as one thing. Across time, something has to carry the first event forward.

A combination: nothing to carry



the combination has to be distinguishable from its parts

Across time: something has to carry



same two events, opposite order, different lesson

Which one came first is part of what is learned.

Module 4's threshold units did combinations too, with the combination fixed in advance.

Classical Conditioning, in a Reflex and in a Worm

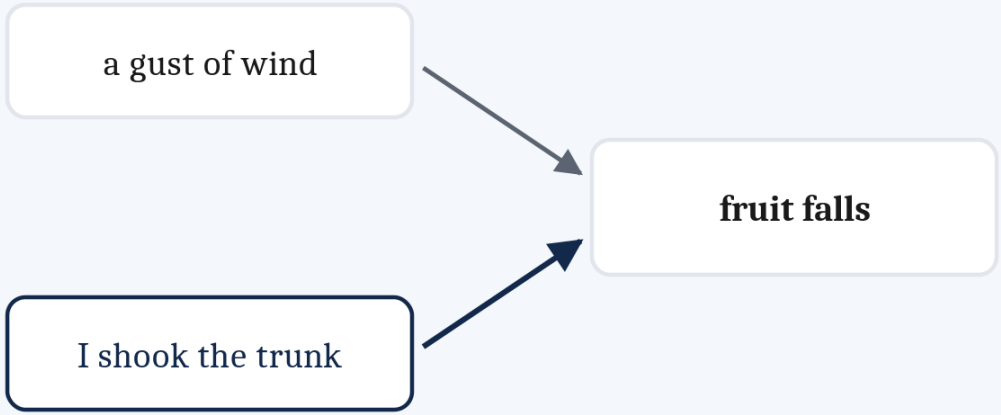
Classical conditioning is learning in which one thing comes to be treated as a sign of another. What changes is which event sets the response off, and nothing about what the animal chooses to do.

Term	The eyeblink case	The worm case
Unconditioned stimulus	A puff of air to the eye	The chemical that causes illness
Unconditioned response	The eye blinks	The malaise
Conditioned stimulus	A tone sounded just before the puff	The food odor paired with the chemical
Conditioned response	The eye blinks to the tone alone	Steering away from the odor

Tier Three: Linking What Happened to the Animal's Own Actions

Two new requirements: telling *this happened* from *this happened after I acted*, and keeping the actions apart. **Operant conditioning**, with its **reinforcer** and **punisher**.

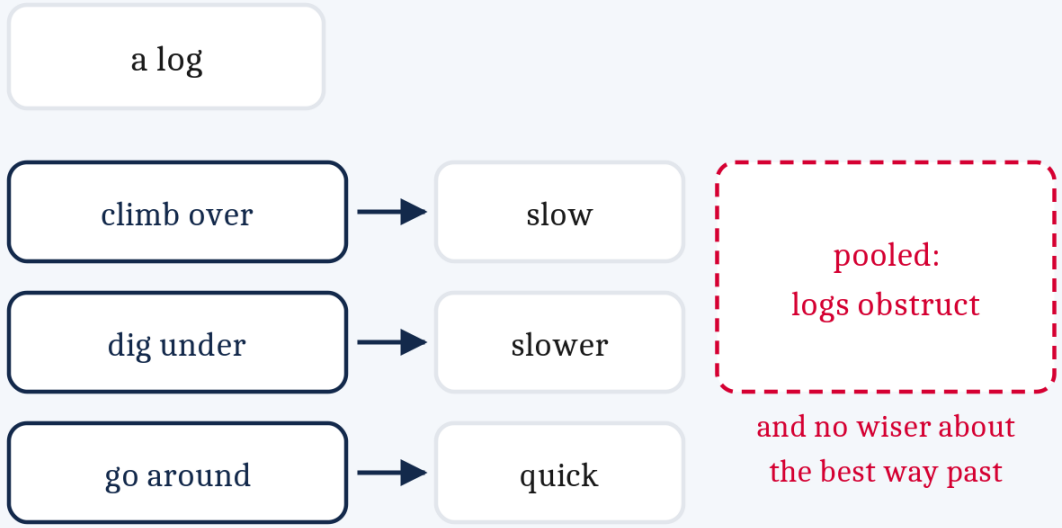
Whose doing was it?



nothing in the falling fruit tells these apart

only an animal with its own shaking alongside the fruit can

Keeping the actions apart



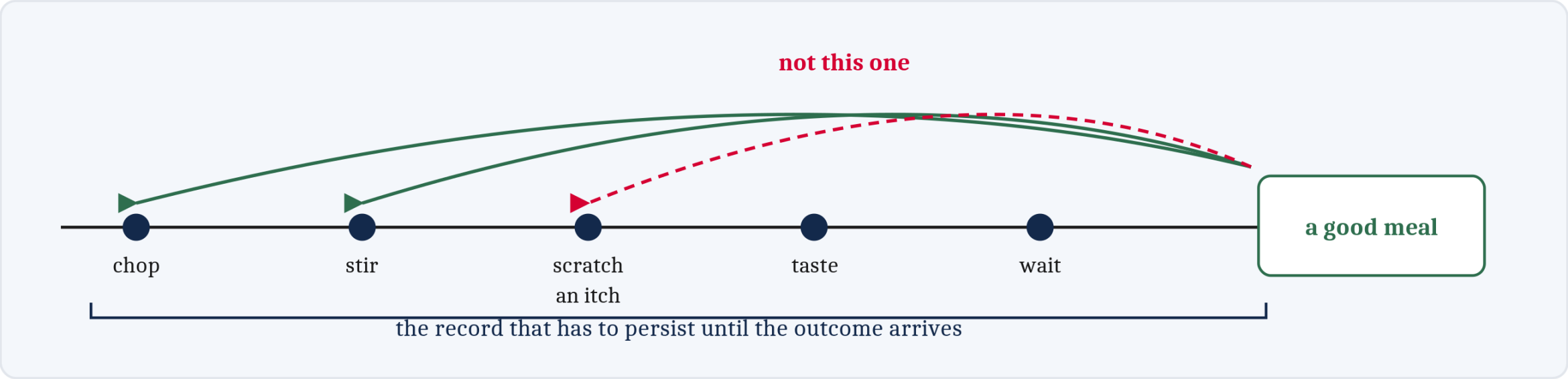
something has to change for one action without changing for the others

A reinforcer makes the response more likely in that situation; a punisher, less likely.

What the animal did is now inside the link, which is what Skinner's word operant names.

Tier Four: Bridging a Delay

Two halves. A **record** of what was done, persisting until the outcome arrives. Then **credit assignment**, apportioning the outcome across everything on the record.



Every action on the record preceded the outcome. Only some of them produced it.

Apportioning the outcome across the record is credit assignment.

Do Animals Without Brains Learn? And How Would We Know?

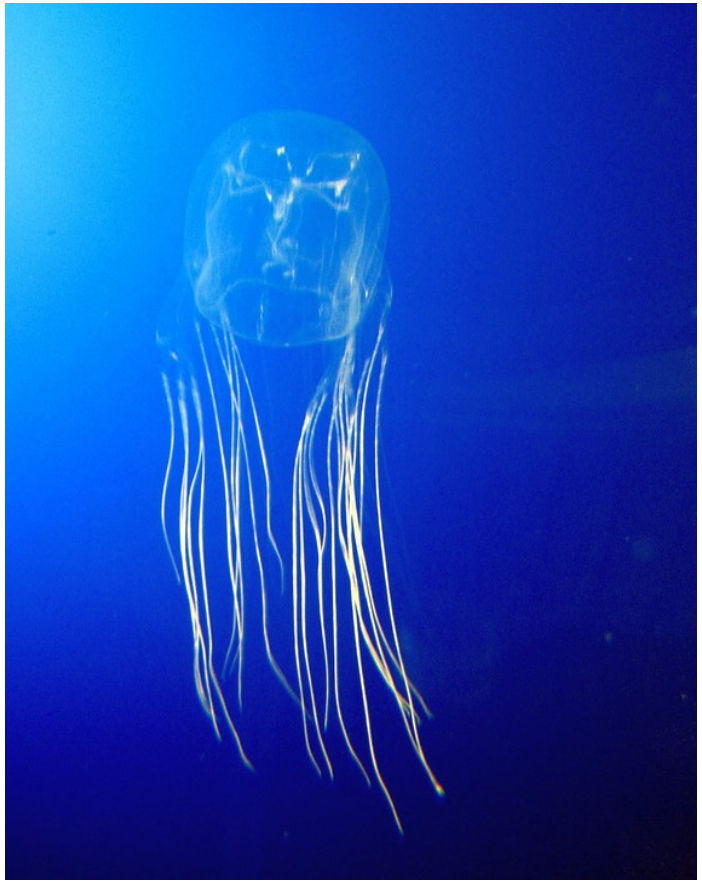
A clean adaptive argument says tier one is all a sessile filter-feeder needs: it does not choose between patches, and does not pursue. The argument predicts an absence, and the absence is not holding up.

Kind of learning	Hydras	Jellyfish	Sea anemones	What would settle the open cells
Habituation (tier one)	Amply shown	Amply shown	Amply shown	Nothing outstanding
Sensitization (tier one)	A hint only	No studies	Shown, with a neurobiological literature	Straightforward replication in the other two groups
Classical conditioning (tier two)	Not studied	Not studied	Two well-controlled demonstrations, one from 1975 and one from 2023	Independent replication in a second laboratory; a retention test beyond ten minutes
Operant conditioning (tier three)	Not studied	One demonstration, 2023, with an isolated ganglion conditioned separately	Suggestive only, on weak controls	Replication in a second species; an acquisition curve and an extinction test

Three Lessons, All of Them Module 2's

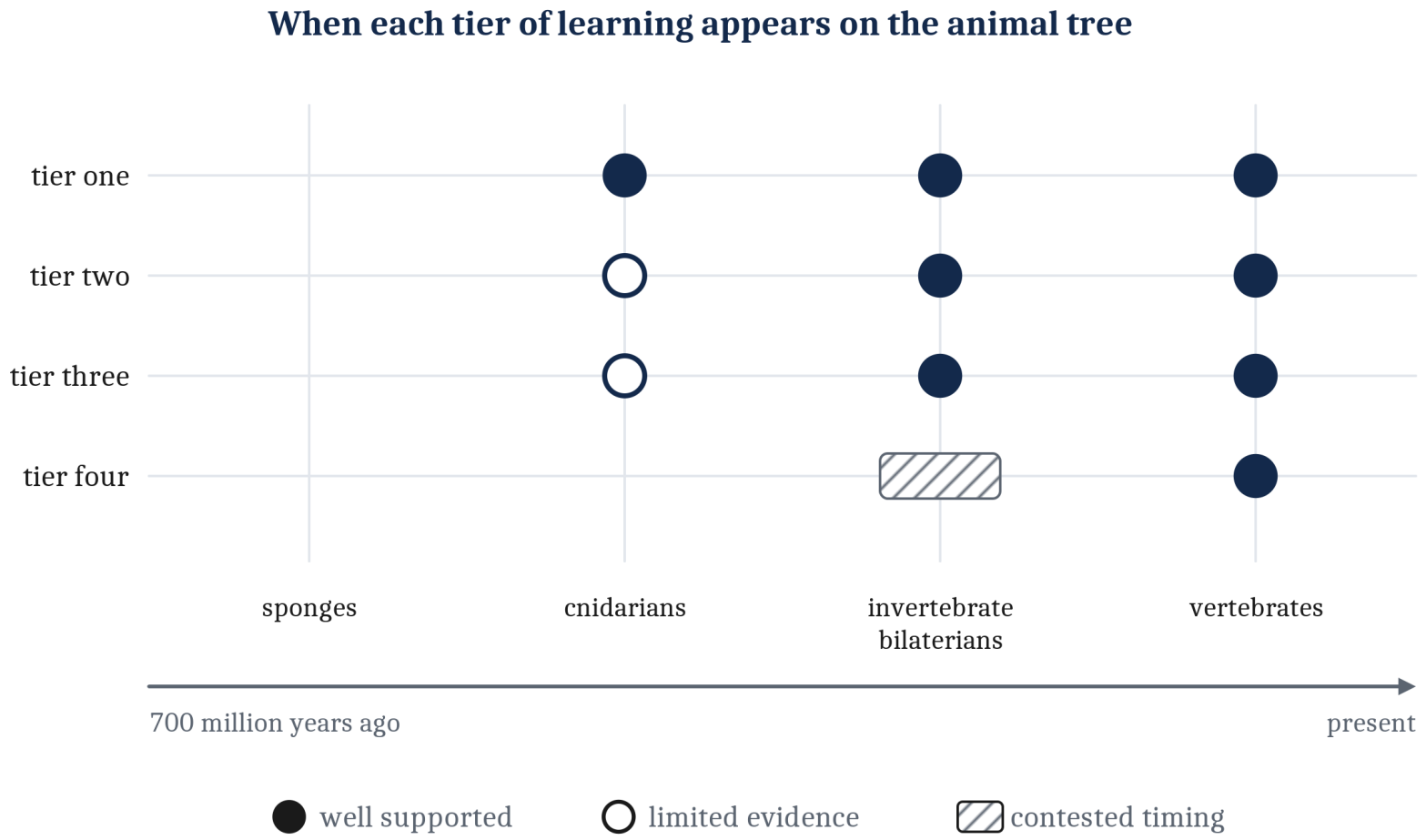
An adaptive argument tested rather than admired. **Morgan's Canon applied asymmetrically** — the same controls in an insect would convince nobody to hesitate. And if cnidarians do learn: homology, or the capacity arrived twice.

The verdict is a shifted burden, not a finding. The question is open.



Most of Learning Is Older Than the Vertebrate

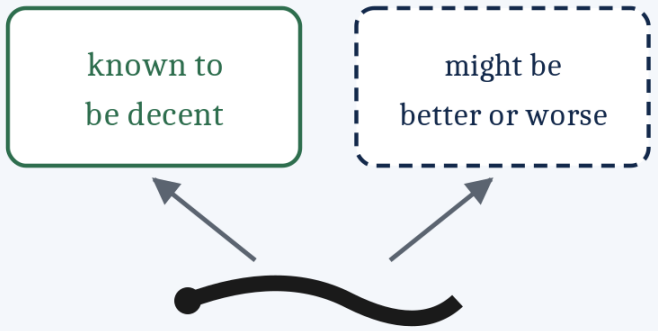
Tiers one and two are old and widespread. Tier three is documented across the bilaterians. **Tier four arrives with vertebrates**: what it adds is the ability to bridge a delay.



Three Problems That Come with Learning

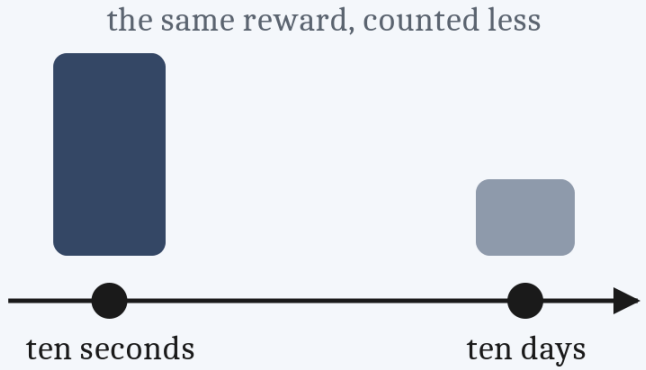
Exploration versus exploitation. How far ahead to care, since ten seconds and ten days cannot count equally. And **stability versus plasticity**, which belongs to every learner.

Exploration versus exploitation



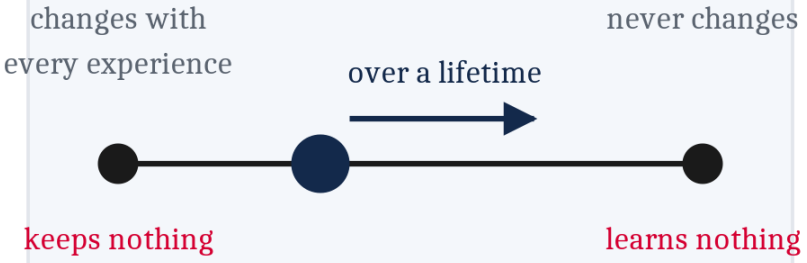
every attempt at one is an attempt
not made at the other

How far ahead to care



some rate of discount is unavoidable

Stability versus plasticity



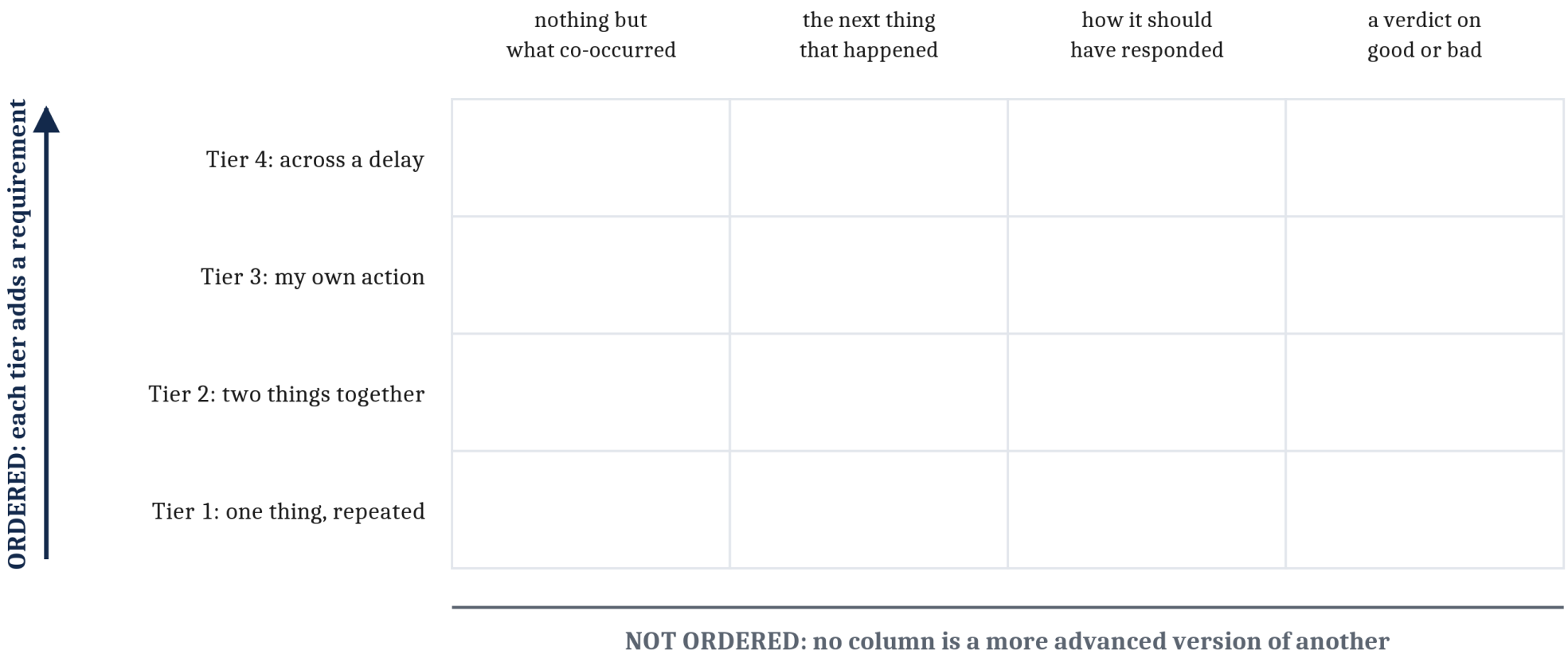
plastic early, more fixed later

The first two belong to an animal learning which actions lead to good outcomes.

The third belongs to every learner, whatever it is learning.

Two Questions, and They Cross Rather Than Stack

The four tiers say what has to be in the link. They say nothing about what the animal has to go on when something changes. **The tiers are ordered. The four things to learn from are not.**



What there is to learn, and what there is to learn from.

Lecture 5.2 divides learning by the columns.

Image Sources (1 of 2)

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6	A contingency the worm never had to learn	fig-copper-contingency.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	One contingency, two routes to a response	fig-two-timescales.png	CC BY 4.0	Jon Willits — drawn for the Module 5 reading · own work
8	What experience modifies	fig-what-changes.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
9	A response declines, and recovers in one step	fig-habituation.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
9	Beadlet anemone, Actinia equina	fig-sea-anemone.jpeg	GFDL 1.2 or later	Attrattorestrano at Italian Wikipedia · https://commons.wikimedia.org/wiki/File:Actinia_equina2.JPG

Image Sources (2 of 2)

Slide	Image	File	Licence	Source
10	A combination, and a link across time	fig-link-across-time.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
12	Whose doing it was, and keeping the actions apart	fig-own-action.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	The record, and apportioning the outcome	fig-credit-assignment.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
15	Box jellyfish, Chironex fleckeri	fig-box-jellyfish.png	CC BY-SA 2.0	Guido Gautsch; derivative work by Mithril · https://commons.wikimedia.org/wiki/File:Avispa_marina_cropped.png
16	The four tiers against the animal tree	fig-tier-timeline.png	CC BY 4.0	Jon Willits — drawn for the Module 5 reading · own work
17	Three problems that come with learning	fig-three-problems.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
18	The tiers are ordered; the columns are not	fig-two-questions-cross.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work